

Mercer International — Quantifying the Transition Layer

A live test of the transition engine against our March-2025 Mercer scenario project (pulp & mass timber). Screening-grade · illustrative figures · Net Zero 2050 unless noted · PV 2025–2050 at 6.5% real.

Bottom line. The Mercer deck scored transition risk qualitatively (severity 1–5) and never priced it. Running Mercer's actual sites through the engine puts a number on it: **~\$3.8B PV of transition cost under Net Zero 2050 (≈7.6% of revenue/yr)** — and it lands in the **same shape** the client survey did: carbon compliance and supply-chain (fibre/transport) are the two dominant channels. Mercer reads as a **low-carbon producer with upside**, not a stranded-asset risk — consistent with the deck's "physical risk > transition risk."

\$3.8B

 NET ZERO 2050 · PV
COST

7.6%

OF REVENUE / YEAR

\$0

STRANDED ASSETS

2

 BUSINESS LINES
MODELLED

Consolidated cost by scenario (firm roll-up)

Scenario	PV transition cost	% rev / yr	L1 carbon compliance	L3 fibre & transport	WACC add-on
Net Zero 2050	\$3.8B	7.6%	\$1.3B	\$1.9B	+10 bp
Delayed Transition	\$2.0B	4.1%	\$0.75B	\$0.93B	+28 bp
Current Policies	\$0.4B	0.8%	\$0.13B	\$0.10B	+18 bp

Note the twist: **Delayed Transition carries the biggest financing penalty (+28 bp)** even though its carbon bill is lower — capital markets price the *disorder*. Net Zero has the highest carbon cost but the mildest premium.

Two business lines behave differently

Line	% rev/yr	WACC
Pulp mills (4 sites)	~8–9%	+14 bp
Solid wood / mass timber	~3.5%	–6 bp

mass timber = transition winner Demand **grows** as it displaces steel & cement; cheaper capital, no stranding. Pulp carries the real compliance + physical exposure.

EU vs North America

Pulp mills	Compliance PV	% rev/yr
EU (Germany)	\$686M	3.2%
N. America (Canada)	\$556M	3.0%

Close, because NGFS puts both in the "advanced" carbon band — the same REMIND data the deck used. The real EU-ETS-vs-Canada gap is a **next-step** refinement.

What the test made us fix (the credibility win)

- **Biogenic wood fibre isn't fossil.** First pass charged Mercer's wood-fibre supply chain at the full fossil carbon price → an absurd 3–4× revenue. Added a **biogenic-Scope-3** carve-out (sustainably-managed fibre is carbon-neutral) → supply-chain cost drops to a defensible level. This is the kind of sector realism the engine needs.
- **New sectors:** pulp & paper (biomass-powered low-carbon producer) and solid wood / mass timber (a beneficiary). Biomass energy excluded from the carbon bill (correct GHG accounting).

Does it match the client work?

Deck's top-scored transition risks were **compliance cost, transportation, biomass cost, wood-supply scarcity**. The engine independently makes **L1 (compliance)** and **L3 (fibre + transport)** the two dominant

channels — same story, now in dollars — and flags the mass-timber **opportunity** the deck called "domestic leadership."

Caveats for the room: figures are illustrative (public FY23–24 reports, allocated across sites), not audited. Screening-grade — not a disclosure number. Emissions are fossil Scope 1+2 only; commodity pass-through applied. Physical risk is a separate tool and, per the original engagement, is the larger story for Mercer.